

Muhammad Usama Naveed

Software Engineer

+92-324-4612107 ♦ Taxila, Pakistan ♦ usamanaveed900@gmail.com ♦ linkedin.com/in/usama-naveed

PROFESSIONAL EXPERIENCE

Results-driven Software Engineer and Data Scientist with a strong background in C++ development, machine learning, and full-stack Python solutions. Experienced in delivering high-impact projects in defense systems, automation, and AI-driven applications, collaborating with international teams, and adhering to strict industry standards such as MIL-STD-498. Adept at developing scalable, modular, and performance-optimized software across domains ranging from artillery fire control systems to machine learning solutions.

Proficient in Qt Creator, Python, PyQt5, Django, Flask, FastAPI, TensorFlow, OpenCV, CAN bus communication, and embedded protocols. Recognized for delivering innovative and precise solutions, with a strong ability to transform projects from concept to functional implementation.

WORK EXPERIENCE

Heavy Industries Taxila (HIT), Taxila, Pakistan

Software Engineer, Feb 2024 – Present

- Collaborating in the development and design of Fire Control Computers for artillery systems, working with cross functional teams from South Africa and Saudi Arabia.
- Completed a one-week advanced external ballistics training in South Africa.
- Developing modular C++ code in Qt Creator for serial, CAN bus, and socket communication.
- Implementing software development processes compliant with MIL-STD-498 standards.
- Designing and implementing data exchange simulators for artillery gun control systems, utilized as test benches.
- Conducting technical evaluations of vendor quotations for off-the-shelf components, including INS, MVR, and Driver Observation Systems and more.

Self-employed — Remote – Taxila, Pakistan

Freelance Software & Machine Learning Developer, Jan 2022 – Dec 2023

- Delivered custom software solutions for international clients via freelancing platforms.
- Developed applications using Python, PyQt5, Django, Flask, Tkinter, FastAPI, and other frameworks.
- Designed and deployed machine learning, data science, and deep learning solutions across various domains.

ZS Ventures, Rawalpindi, Pakistan

Python Developer, Apr 2021 – Oct 2021

- Developed e-commerce platforms using the Django framework.
- Created and integrated trained ML models into production environments.
- Built custom admin panels for managing e-commerce operations.

TereSol, Islamabad, Pakistan

Machine Learning Engineer – Intern, Jul 2019 – Sep 2019

- Implemented image and video processing solutions using OpenCV, including facial and eye detection with AR overlays.
- Designed interactive Python UIs for form-based data entry, CSV storage, and real-time video streaming.
- Integrated TensorFlow/Keras-trained object detection and counting models into live video streams.

- Worked on RTSP IP camera integration and developed multi-video streaming using a server–client architecture with ImageZMQ.

AKSA-SDS, Islamabad, Pakistan

Embedded Systems – Intern, Jul 2017 – Sept 2017

- Developed TCP protocol-based communication systems in C++.
- Implemented serial port data transmission and reception for embedded applications.

EDUCATION

UNIVERSITY OF ENGINEERING AND TECHNOLOGY, TAXILA (*2023 – Present*)

- Master in Data Science

BAHRIA UNIVERSITY, ISLAMABAD (*2016–2020*)

- Bachelors of Software Engineering

COURSES

iNkwazi Software Solutions, Centurion, South Africa (*16 Sep 2024 – 20 Sep 2024*)

- Advanced External Ballistics - Short Course

PUBLICATIONS

- **Naveed, Muhammad Usama**, Muhammad Munwar Iqbal, Saqib Majeed, Farooq Ali, and Qamas Gul Khan Safi. "Lung Cancer Classification through Transfer Learning and Deep Feature Extraction using EfficientNetB3." *Journal of Computing & Biomedical Informatics* 9, no. 02 (2025).

LANGUAGES

- English (Professional Proficiency)
- Urdu (Mother Tongue)